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Attention Entanglement

Simultaneous query and semantics updating

Gary Nan Tie, Sep 25, 2025

Abstract
Transformer attention uses statistical correlation to update queries, and fibration abduction attention on top, takes into account key-value semantics, finding a key to explain the value returned. The latter has a quantum interpretation, indeed a fibration lifting is an entanglement. Entanglements (inner loop) can be refined using a Renyi α-entropy criterion, as part of a simultaneous query and key update (outer loop). Thus the value returned then reflects refinement of both query-key similarity and key-value semantics.

Attention Entanglement

Begin Outer Loop - <u>entanglement $\sigma(p,m)$</u> :

Input $(k_j, v_j) \in K \times V$, $j \in [m]$

$(q_i, q_i') \in Q \times Q'$, $i \in [n]$

Learn NN $p : K \to V$ such that $p(k_j) =_\varepsilon v_j$, $j \in [m]$

and NN $m : Q \to Q'$ such that $m(q_i) =_\varepsilon q_i'$, $i \in [n]$.

Given p and m, learn $\sigma : Q \otimes V \to Q' \otimes K$ such that

$$\begin{tikzcd}
Q \otimes K \arrow[r, "m \otimes 1"] \arrow[d, "1 \otimes p"'] & Q' \otimes K \arrow[d, "1 \otimes p"] \\
Q \otimes V \arrow[r, "m \otimes 1"'] \arrow[ur, dashed, "\sigma"] & Q' \otimes V
\end{tikzcd}$$

commutes up to $=_\varepsilon$, that is:

i) $\sigma \begin{bmatrix} q_i \\ p(k_j) \end{bmatrix} =_\varepsilon \begin{bmatrix} m(q_i) \\ k_j \end{bmatrix}$,

See [20].

ii) $\pi_1 \sigma \begin{bmatrix} q_i \\ v_j \end{bmatrix} =_\varepsilon m(q_i)$,

iii) $p \, \pi_2 \sigma \begin{bmatrix} q_i \\ v_j \end{bmatrix} =_\varepsilon v_j$, $i \in [n]$, $j \in [m]$.

Renyi α-entropy

For $0 < \alpha < \infty$, $\alpha \neq 1$,

$$H_\alpha(\{p_i\}_{i=1}^n) \triangleq \frac{1}{1-\alpha} \ln \left(\sum_{i=1}^n p_i^\alpha \right)$$

where $0 \leq p_i \leq 1$ and $\sum_{i=1}^n p_i = 1$.

For $\alpha = 0, 1, \infty$, $H_\alpha \triangleq \lim_{\gamma \to \alpha} H_\gamma$.

Fix $i \in [n]$, let $\delta_{ij} = \underset{j}{\text{softmax}}\ \text{score}(q_i', k_j)$

and $j^* = \arg\max_j \delta_{ij}$, where score is a similarity function.

$\{\delta_{ij}\}_{j=1}^m$ is a distribution over $\{k_j\}_{j=1}^m$.

Fix $j \in [m]$, let $\gamma_{ij} = \underset{i}{\text{softmax}}\ \text{score}(q_i', k_j)$

and $i^* = \arg\max_i \gamma_{ij}$.

$\{\gamma_{ij}\}_{i=1}^n$ is a distribution over $\{q_i'\}_{i=1}^n$

Choose hyperparameter $\alpha \geq 0$.

Define $E_\sigma \triangleq \frac{1}{n} \sum_{i=1}^{n} H_\alpha^i$,

where $H_\alpha^i \triangleq H_\alpha(\{\delta_{ij}\}_{j=1}^m)$.

E_σ is an entropy measure of entanglement σ,

the lower, the greater the certainty.

Begin Inner Loop - <u>entanglement learning</u>

Apply σ to update q_i' and k_j:

$$\begin{bmatrix} q_i' \\ v_j \end{bmatrix} \xmapsto{\sigma} \begin{bmatrix} \vec{q}_{ij} \\ \vec{k}_{ij} \end{bmatrix}, \quad \vec{q}_{ij} \triangleq \pi_1 \sigma \begin{bmatrix} q_i' \\ v_j \end{bmatrix} \in Q'$$

$$\vec{k}_{ij} \triangleq \pi_2 \sigma \begin{bmatrix} q_i' \\ v_j \end{bmatrix} \in K$$

On data $(q_i, \vec{q}_{ij*})$, $i \in [n]$, learn $\vec{m} : Q \to Q'$.

On data (k_{i^*j}, v_j), $j \in [m]$, learn $\vec{p} : K \to V$.

From $\vec{p}$ and $\vec{m}$, learn $\vec{\sigma} : Q \otimes V \to Q' \otimes K$.

If $E_{\vec{\sigma}} < E_\sigma$, continue refining entanglements,

vis-à-vi apply $\vec{\sigma}$ to update $\vec{q}_{ij*}$ and $\vec{k}_{i^*j}$, and so on...

End Inner Loop

Return $\vec{\sigma} : Q \otimes V \to Q' \otimes K$

when $E_{\vec{\sigma}}$ is suffiently low, otherwise σ..

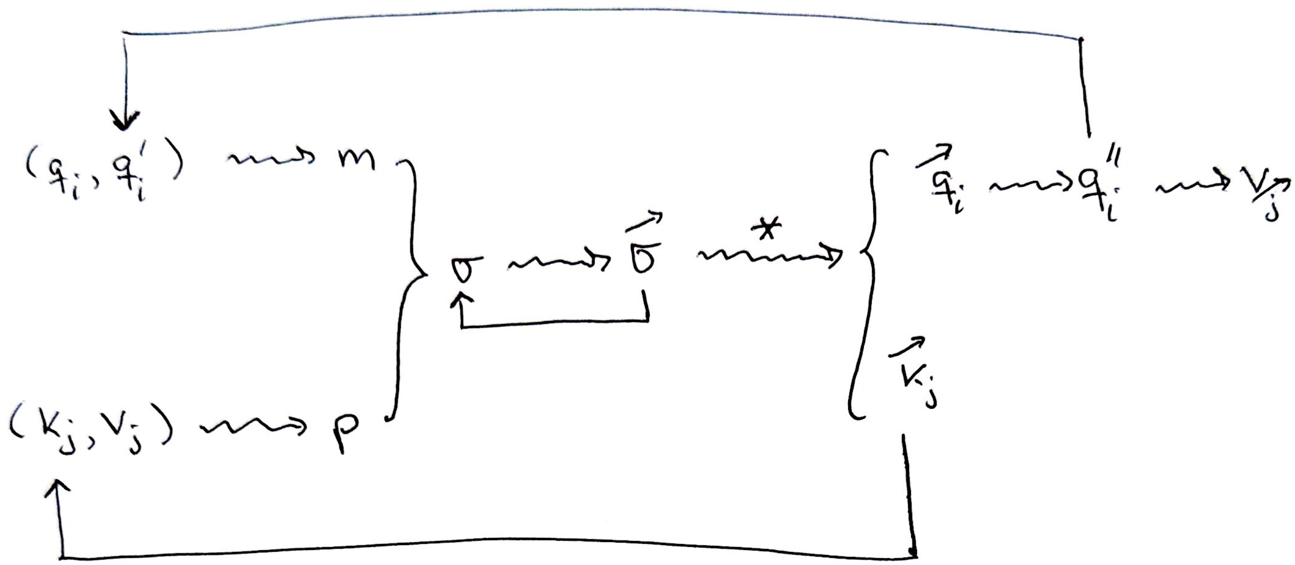
—//—

* To be able to reuse previous notation, now replace σ with $\vec{\sigma}$ just returned. In which case, for $i \in [n]$, define $q''_i = \sum_{j=1}^{m} \beta_{ij} v_j$, where $\beta_{ij} \triangleq \operatorname{softmax}_j \text{ score}(\vec{q}_{ij^*}, v_j)$ and define $\vec{j} = \arg\max_j \beta_{ij}$.

In response to query q'_i, return $q''_i \in Q'$ and $v_{\vec{j}} \in V$.

End Outer Loop

Replace* q'_i with q''_i and k_j with $\vec{k}_{i^*j}$, and repeat this nested loop until perplexity is acceptable.



Updating Schematic:

References (most recent first)

[20] 'Quantum Attention'
Gary Nan Tie, Sep 19, 2025
DOI: 10.13140/RG.2.2.19514.25289
The quantum nature of fibration abduction attention.
A LLM paradigm shift from similarity to explanation.

[19] 'Fibration Attention Dynamics'
Gary Nan Tie, Sep 13, 2025
DOI: 10.13140/RG.2.2.36796.09600
We characterize the dynamics of query updating coming from fibration abductive attention mechanisms, using perturbed Koopman operators that are invertible. Importantly, we identify conditions for Lyapunov stability of solutions near equilibrium points. Depending on the Koopman resolvent, some key-value pairings can have query update trajectories that are Lyapunov stable and converge.

[18] 'Jury Attention'
Gary Nan Tie, Sep 10, 2025
DOI: 10.13140/RG.2.2.25750.00322
To address the Rashomon effect when multiple key-value semantics are present, we introduce a jury attention mechanism via fibrations. This accomplishes query updates that take into account differing world views.

[17] 'Attention Legos'
Gary Nan Tie, Sep 8, 2025
DOI: 10.13140/RG.2.2.33398.05447
Attention building blocks for machines to learn world models?

[16] 'Deep Fibration Attention'
Gary Nan Tie, Aug 31, 2025
DOI: 10.13140/RG.2.2.17946.30402
In fibration attention, a lifting simultaneously updates a query and abductively finds a key to explain a value. This all depends on the key-value pairing (k,v). By determining other keys that are highly similar to k, to also explain v, we can enhance key-value semantics to achieve more meaningful attention. Moreover, we can recursively layer keys upon keys in a tree, to deepen the set of value explanations.

[15] ‘Contextual Fibration Attention’
Gary Nan Tie, Aug 29,2025
DOI: 10.13140/RG.2.2.12205.35044
In a query-key-value attention model, the key-value pairing conveys language semantics. Query context also carries meaning.
We introduce an abductive fibration attention model that holistically captures both. A fibration lifting, jointly learns from query context and language semantics (that abductively finds a key to best explain the returned value).

[14] ‘Multi-head abductive attention’
Gary Nan Tie, Aug 25, 2025
DOI: 10.13140/RG.2.2.20597.23521
Transformer multi-head attention can focus on different parts of an input query sequence, however without regard to potentially different key-value semantics. This new primitive for formal query updating, constructs updates to a query in parallel, by learning multi-head abductive fibration liftings, to refine similarity based attention, utilizing the semantics of multiple key-value pairings.

[13] ‘Formal query updating to generate problem specific fibration abduction attention algorithms’
Gary Nan Tie, Aug 23, 2025
DOI: 10.13140/RG.2.2.33029.41446
A formalism to construct problem specific query update algorithms founded on standard attention and fibration attention primitives.

[12] ‘Fibformer - a fibration transformer’
Gary Nan Tie, Aug 22, 2025
DOI: 10.13140/RG.2.2.36293.10723
We introduce a new transformer with novel fibration attention based on abductive reasoning. This coherent approach to query updating enables controlled structured exploration of continuation values.

[11] 'LLM Bonsai - pruning FibChat
Gary Nan Tie, Jul 25, 2025
DOI: 10.13140/RG.2.2.21984.80644
Each step in learning the diagonal lifting of a fibration, used to simultaneously update a query and find a section of the key-value pairing, represents an opportunity for intervention to direct the growth of our query refinement tree by selective pruning, so as to avoid hallucinations, LLM bonsai!

[10] 'FibChat - It makes stuff up, but with structure.'
Gary Nan Tie · July 2025
DOI: 10.13140/RG.2.2.10549.59364/2
FibChat is a LLM based on fibration attention, that enables coherent structured mitigation of hallucinations.

[9] 'Attention Fibration Abduction'
Gary Nan Tie, Jul 2, 2025
DOI: 10.13140/RG.2.2.29060.23680/1
Transformer attention and abductive reasoning are intimately connected, embodied by fibrations. Suitable attention mechanisms can make a key-value pairing a learnable neural network fibration for abductive reasoning. A fibration lifting explains an attention output value to a query, by identifying the key paired to it, explainable attention through abduction.

[8] 'Parsimonious Neural Network Abduction'
Gary Nan Tie, Jun 22, 2025
DOI: 10.13140/RG.2.2.15695.80804
For hypotheses whose effect is manifested by observations, abduction seeks to explain a given observation by finding a hypothesis whose effect is that observation. Abductive reasoning has a fibration semantics that can be implemented by neural networks; a step towards artificial general intelligence. In this note we introduce a parsimonious choice of explanation depending on one's utility function.

[7] ‘Neural Nerwork Abduction’
Gary Nan Tie, Jun 13, 2025
DOI: 10.13140/RG.2.2.18506.07360/1
For hypotheses whose effect is manifested by observations, abduction seeks to explain a given observation by finding a hypothesis whose effect is that observation. Abductive reasoning has a fibration semantics that can be implemented by neural networks; a step towards artificial general intelligence.
A natural application is in making a medical diagnosis. Fibration structure and properties enable coherency and consistency in this process. Moreover being machine learnable, can help physicians uncover diagnostic insights in large datasets.

[6] ‘Meta-Abduction’
Gary Nan Tie, Jun 10, 2025
DOI: 10.13140/RG.2.2.22022.64323
We note that for fibration semantics of abductive reasoning, abduction of attention is a meta-effect, not section reversal per se.

[5] ‘Abductive Inference via Explainable Neural Networks’
Gary Nan Tie, May 25, 2025
DOI: 10.13140/RG.2.2.20953.84320
Explainable abductive inference by invertible neural networks is examined in the context of fibration semantics.

[4] ‘Abductive Reasoning Precedence’
Gary Nan Tie, May 23, 2025
DOI: 10.13140/RG.2.2.19913.45923
Fibration liftings used to model abduction are not necessarily unique. So chains of abductive reasoning are a series of certain choices of liftings. To keep track of this precedence we introduce a simple mechanism to mark liftings. This transparency enables one to analyze decision points in an exploit-explore strategy for optimizing chains of abductive reasoning.

[3] ‘Coherent Abductive Medical Diagnosis’
Gary Nan Tie, May 15, 2025
DOI: 10.13140/RG.2.2.16148.41605
Disease causes symptoms. In the other direction, given a patient's symptoms, medical diagnosis seeks diseases that explain the observed symptoms, a form of abductive reasoning. Abduction can be viewed as a fibration, a mathematical model, which when applied to clinical decision making enables coherent, transparent, refinable medical diagnoses. Moreover, given enough data, fibrations are potentially machine learnable, providing an interactive AI learning tool for physicians to better understand and refine their diagnostic thinking.

[2] ‘Fibration Abductive Learning’
Gary Nan Tie, Mar 4, 2025
DOI: 10.13140/RG.2.2.27284.82562/2
Supplement DOI: 10.13140/RG.2.2.24804.28800/1
Neural networks like MLP (multilayer perceptrons) and KAN (Kolmogorov-Arnold networks) can solve, by induction, regression, classification and time series problems. Deduction is used by neural networks in rule based systems, logic programming, automated theorem proving and reinforcement learning. This naturally begs the question about applying the third kind of inference abduction, seeking the simplest and most likely explanation for a set of observations. To conceptualize and perform abductive machine learning, categorical fibrations are proposed, as they coherently give both context and structure for abductive reasoning via a tautological attention mechanism.

[1] ‘Fibrations explain all you need!
- Fibrations, Abduction, Attention
Gary Nan Tie, Mar 4, 2025
DOI: 10.13140/RG.2.2.35984.52488
To conceptualize and perform abductive machine learning, categorical fibrations are proposed, as they coherently give both context and structure for abductive reasoning via an attention mechanism.
Moreover, 2-categorical lifting compatibility conditions ensure consistent hierarchical and parallel explanations.